

(a) #1) Stratified sampling gives us test distribution closer to the overall distribution as compared to the random sampling.
TRUE

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#2) A standard correlation coefficient value of 0.98 represents strong positive correlation between features
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#3) Why is it a good idea of to create a copy of the training set?

So that we can freely manipulate it without altering original training dataset

#4) Write the purposes of matplotlib & seaborn.

* Matplotlib: Used for data visualization

It is a python library, a popular 2D plotting library for Python that produces high quality charts and figures in various formats.

* Seaborn: Used for data visualization

It is a statistical data visualization library based on matplotlib.

It provides a high-level ~~and~~ interface for drawing attractive & informative statistical graphics.

#5) Define sklearn

sklearn is an open source Python library that implements range of machine learning, preprocessing, cross validation, visualization algorithms using an unified interface.

#6) What is a way to use heatmap?

A heatmap can be used to visualize correlation matrix.

#7) The function `isna()` is used for which of the following purposes? to detect missing (NaN) values

#8) Which of the following methods may be used to drop rows containing missing values?
`drop()` `dropna()`

#9) Sklearn provides SimpleImputer class for filling up missing values
"TRUE"

#10) fit_transform() is method on OrdinalEncoder object is called for which purpose? : converts TEXT to NUMBERS

#11) * Scikit-learn provides a pipeline class to line up transformations in an intended order.

#12) Suggest a good evaluation measure for regression models.
• Mean Squared Error

#13) Cross validation provides a separate MSE for each validation set.

#14) Ensemble learning ~~Build~~ Building a model on top of another ~~model~~ model.

#15) What is the remedy for underfitting?
to have less constraints / regularizations

#16) Cross Validation?

Cross validation, is a statistical technique used in machine learning and statistics to assess the performance of a predictive model.

Its primary purpose is to evaluate a model's ability to generalize new, unseen data

Most common form of CV is K-fold CV

Steps:

1. Dataset Splitting : Splits the dataset into K approximately equal-sized folds (subsets)

2. Model training and testing

Model training on (K-1) folds

Test Data is the remaining 1-fold.

3. Performance Metric Calculation

A performance metric (eg: accuracy, mse) is calculated on the test set.

4. Iteration : Steps (2), (3) are repeated K-times, with a different fold as the test set in each iteration.

5. Performance Summary : The performance ^{score} ~~metric~~ in each iter are averaged to provide overall assessment of model's performance

* Benefits of cross validation

- **Reduced Variance**: Cross validation helps to reduce the variance in performance estimates by using multiple test sets.
- **Better Generalization**: Provides a more reliable better estimate of how well a model will perform on unseen data.
- **Optimal Use of Data**: Allows the model to test and train on data all available data; maximizing the use of the dataset.

Q#17) What are hyperparameters?

Hyperparameters: Hyperparameters are the configuration settings of a model that are external to the model itself and are not ~~learned~~ learned from the data.

Eg.

- 1) **Learning Rate** (Gradient based optimization algo)
- 2) **Batch Size** (Gradient based optimization algo)
 - determines no of samples used in each iteration of training
 - determines the step size in updating the model weights during training.
- 3) **Number of trees** (for ensemble methods like RandomForest) specifies number of individual decision trees in the ensemble.
- 4) **Depth of trees** (for decision tree-based models) Sets maximum depth of the individual decision trees.
- 5) **Regularization Parameters** (α in Lasso / Ridge Regression) controls amount of regularization in the model.
- 6) **Kernel parameters** (C , gamma in Support VM) influence the shape and ~~size~~ complexity of decision boundaries.
- 7) **Number of neighbors** (for k-nearest neighbors) specifies no of neighbors used in the prediction.

GridSearchCV is a method ~~that~~ in sklearn that ~~provides~~ performs an exhaustive search over a specified parameter grid, evaluating the performing of a machine learning model with different hyperparameter combinations.

Commonly used for hyperparameter tuning

importing required libraries

from sklearn.model_selection import GridSearchCV

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestRegressor

from sklearn.metrics import mean_squared_error

splitting the data

X_train, X_test, y_train, y_test = ~~the~~ train_test_split(X, y,
test_size = 0.2, random_state = 42)

model object creation

model = RandomForestRegressor()

defining parameter grid (dictionary of parameters : list of values)

param_grid = {

'n_estimators': [10, 50, 100],

'max_depth': [None, 10, 20],

'min_samples_split': [2, 5, 10],

'min_samples_leaf': [1, 2, 4]

}

creating gridsearchcv object

grid_search = GridSearchCV(estimator = model, param_grid =
param_grid, scoring = 'neg-mean-squared-error',
cv = 5)

fitting train data

grid_search.fit(X_train, y_train)

retrieving set of best params

best_params = grid_search.best_params_

best_model = grid_search.best_estimator_ # best model
(with best hyp. comb.)

predicting the output from X_test

y_pred = best_model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)

print(f'Mean Squared error on test set : {mse}')

evaluating the model using mse

Q#19/ What is RandomizedSearchCV?

RandomizedSearchCV is exactly similar to GridSearchCV except for the fact that for a given set of param-grid, any randomly combination of the hyperparameter gets used in turns for RandomizedSearchCV.

Q#20/ What is fit_transform()?

fit_transform() is a method associated with transformers, which are classes that implement the 'fit()' and 'transform()' methods. It both fits the data and applies the learned transformations to the input data.

#21/ What are some different kinds of sklearn objects?

Estimators

Predictors

Transformers

#22/ What does 'sklearn.linear-model' implement?

It implements classical algorithms like linear regression, logistic Regression.

#23/ Give an example of supervised learning problem.
A bank looking to decide the amount of personal loans to its customers based on transactions in their amount.

Q#24/ State two examples of regression task.

1) A bank deciding loan amt for cust.

2) Predicting rainfall this year based on past record.

Q#25/ L1 regularization $\rightarrow$ Lasso

L2 regularization $\rightarrow$ Ridge

Q#26/ Why is it good/necessary to split data into train test?

To ensure low variance & better performance on unseen data.

Q#27) Which function is fully used for describing the dataset?
• DESCR

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Q#28) Which of the following statements are true for openml.org?
(Only true statements are noted below)

- Public repository for machine learning data & experiments
- openml.org allows everybody to upload open datasets

* make_blobs generates random data for clustering
↳ present in scikit learn sklearn.datasets

$X, y = \text{make_blobs}(n\text{-samples}=300, \text{centers}=3, \text{random-state}=42)$

* Feature matrix shape (#samples, #features)

Target matrix shape (#samples, #labels)

Q#29) What are different label classes in 'Iris' dataset?
['setosa', 'virginica', 'versicolor']

Q#30)

Given some dataset (a dictionary) $\text{data} = [\{ \text{'#courses': 4, 'GPA': 6.7} \}, \{ \text{'#courses': 4, 'GPA': 8.5} \}, \dots]$

How to extract the features?

Use DictVectorizer from sklearn.feature-extraction

↳ extracts info in sparse matrix or normal matrix

from sklearn.feature-extraction import DictVectorizer

$dv = \text{DictVectorizer}(\text{sparse}=\text{False})$ // object creation

$\text{data} = dv.\text{fit_transform}(\text{data})$

Q#31) For StandardScaler() transformation what is the formula?

$$x' = \frac{x - \mu}{\sigma} \quad \text{where} \quad \sigma^2 = \frac{1}{N} \left(\sum (x_i - \mu)^2 \right)$$

Q#32) Formula for MinMaxScaler()

$$x' = \frac{x - x.\text{min}}{x.\text{max} - x.\text{min}}$$

Q#32) Formula for `maxAbsScaler()`.

$$m = \max(\max(|x_i|), |x_{i,\max}|) \text{ then } x = \frac{1}{m} \cdot x$$

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Q#34) `FunctionTransformer()` ?

from `sklearn.preprocessing` import `FunctionTransformer`
used to apply a function of choice to the data values

import `numpy` as `np`

`X = np.array([[1], [2], [3]])`

`ft = FunctionTransformer(np.mean)`

`ft.fit_transform(X)`

function of
choice

Q#35) Consider $X = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 3 \\ 1 \end{bmatrix}$. Convert X using `OneHotEncoder()`

$$X = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 3 \\ 1 \end{bmatrix} \xrightarrow[\text{matrix}]{\text{sparse}} X' = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad 5 \times 4$$

Q#36) Convert categorical feature X to X' using `MultiLabelBinarizer()`

$$X = \begin{bmatrix} \{ \text{'Heart Disease'} \} & \{ \text{'Diabetes'} \} \\ \{ \text{'High BP'} \} & \\ \{ \text{'Heart Disease'} \} & \{ \text{'High BP'} \} \\ \{ \text{'Diabetes'} \} & \end{bmatrix}$$

Heart disease

2nd column Diabetes

High BP

$$X' = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Q#37) What does `LabelBinarizer()` do?

converts multi-class labels to binary labels

Q#38) What does `add_dummy_feature()` do?

Available in `sklearn.preprocessing` all set to
Adds a dummy feature column with default value 1

Q#39) `SimpleImputer` apply to $X = \begin{bmatrix} [3, 8] & [4, 1] & [8, \text{np.nan}] \end{bmatrix}$

`imp = SimpleImputer(missing_values = np.nan, strategy = 'mean')`

$$X' = \frac{8+4}{2} = 6$$

Q#40) Which of the following are API for feature selection based on filter based methods? ☒ Variance Threshold ☒ Select Percentile

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Q#41) What does "API VarianceThreshold" do?

* It removes all features with variance below a certain threshold value from input feature matrix.

Q#42) What task does the API "SelectKBest" do?

* Removes all but the K-highest scoring features.

Q#43) What will the following code implement?

```
transformer = GenericUnivariateSelect(chi2,  
                                     mode='k-best',  
                                     param=20)  
X_new = transformer.fit_transform(X, y):
```

Selects top 20 best features based on χ^2 scoring function.

Q#44) When can χ^2 be used as a scoring function?

χ^2 can only be used in classification problems.

Q#45) When can F-statistics be used as a scoring problem?

F-statistics ^{can be} used as a scoring function for both regression & classification problems.

Q#46) Define RFE (Recursive Feature Selection)

- * It is a wrapper based feature selection.
- * RFE works by searching for a subset of features by starting with all features in the training dataset and successively removing features until the desired number remains.

Q#47) The argument of the parameter threshold in the class

SelectFromModel can be integer
'mean'
'median'

Q#48) Which approach is used in the class SequentialFeatureSelection()?

Forward selection approach.
Backward selection approach.

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Q#49) Which module in sklearn deals with applying transformation on subset of diverse features and combine them?

`sklearn.compose` : Transformation.

Q#50) Which module in sklearn, is used for dimensionality reduction techniques?

`sklearn.decomposition`

Q#51) Which are the correct statements about PCA?

- It is linear dimensionality reduction technique.
- It uses SVD to project feature matrix or data to a lower dimensional space.
- The ~~separate~~ subsequent PCs are orthogonal to the first PC.

Q#52) What task does the module sklearn.pipeline do?

~~Provides~~ Provides utilities to build a composite estimator, as a chain of transformers & estimators.

Q#52) What are two ways of creating a pipeline object?

`Pipeline()` and `make_pipeline()`.

Q#53) ?usage of sklearn.pipeline.FeatureUnion :

- It concatenates results of multiple transformer objects.
- The list of transformer objects can be executed in parallel.

Q#54) The purpose of a DictVectorizer object is to :

convert lists of mappings of feature name and feature value into a matrix.

Q#55/ Which of the following options are data preprocessing techniques?
(only correct ones are noted)

- ✓ Handling categorical features
- ✓ Handling imbalanced data
- ✓ Adding polynomial features

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* TREE algo are not affected by feature-scaling

Q#56/ Which of the following are true about (KDE kernel density estimation)

- ✓ KDE has a non-parametric way of operation
- ✓ KDE is used to estimate the probability density function of a random variable
- ✓ KDE uses Gaussian kernels
- ✓ KDE includes automatic bandwidth determination

Q#57/ What is true about violinplots?

- ✓ Violin plot shows the probability density of the data at different values, smoothed by a kernel density estimator
- ✓ Violin plot shows the distribution of quantitative data across several levels of one (or more) categorical variables such as that those distributions can be compared.
- ✓ Violin plot features a density kernel ~~estimator~~ estimation of the underlying distribution.

Q#58/ What will the following code implement?

```
from sklearn.preprocessing import FunctionTransformer  
ft = FunctionTransformer(np.log2)
```

Q#59/ What 'discretization' do?

It provides a way to partition continuous features into discrete values

Q#60/ Where does TransformedTargetRegressor reside in sklearn library?

sklearn.compose

Q#61) What is the output of the following code?

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```
import numpy as np
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.feature_selection import SelectKBest, mutual_info_regression

# download data
X_cal, y_cal = fetch_california_housing(return_X_y=True)

X, y = X_cal[:2000, :], y_cal[:2000]
skb = SelectKBest(mutual_info_regression, k=4)
X_new = skb.fit_transform(X)
X_new.shape
```

Shape of feature matrix after feature selection is (2000, 4)

Q#62) What's the difference between SelectPercentile and SelectKBest

SelectPercentile selects top k percentile of all features

SelectKBest selects top k numbers of highest scoring features based on a function.

Q#63) Describe Pipeline

- ✓ Pipelines sequentially apply a list of transformers
- ✓ Intermediate steps of pipeline must be transformers
- ✓ Purpose of pipeline is to assemble several steps that can be cross validated together, while setting different parameters.
- ✓ Parameters of the estimators in the pipeline can be accessed using the `-syntax-estimator-parameter-`

WB

Q#64) What are the valid strategy parameter in Drum Regression?
mean
median
quantile

Q#65) Which object can be used to solve linear regression using iterative optimization? LR = SGDRegressor

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Q#66) For performing Linear Regression using sklearn, which of following module will be appropriate?

sklearn.linear_model

Q#67) Which sklearn model is used to do necessary transformations on the training data before performing polynomial regression?

sklearn.preprocessing

- Q#68)
- Generalization Error : Test Error
 - Empirical Error : Training error

Q#69) Which sklearn module is used for model evaluation & scoring

sklearn.metrics

Q#70) Which evaluation metric is best suited for targets with exponential growth?

mean squared log error

Q#71) In which sklearn modules does shuffle split function lie?

sklearn.model_selection

used with 'n_splits=5' to generate 5

different random train-test

splits { ~~index~~ 5 so, 5 ~~test~~ indexing split store as tuple

The 'split' method is then used to obtain the shuffled indices for each split.

tuple (train-index, test-index)

Q#72) Which sklearn module does permutation-test-score function lie?

sklearn.model_selection

Q#73) Which sklearn module contain Dummy Regression?

~~sklearn.module~~ sklearn.dummy

Baseline regression model

Q#74) What is the pipeline API of sklearn useful for?
for combining datapreprocessing & model building

Q#75) Which libraries are used for inplace data manipulation in python?
✓ numpy
✓ pandas

Q#76) Which of the following python modules provide utilities for data visualization?
matplotlib, pandas, seaborn

Q#77) Which module implement cross validation routines?
sklearn.model-selection

Q#78) What cross validation routine do we use when we desire the complete details of the model trained on each validation subset of the training set?

cross-validate

Q#79) For performing Stochastic Gradient Descent which of the following sklearn module will be appropriate?

sklearn.linear-model

Q#80) Which sklearn module contain mean-absolute-error

sklearn.metrics

Q#81) Which sklearn module contain validation curve funcⁿ lie?

sklearn.model-selection

Q#82) What quantity does warm-start = True option in SGDRegressor tweak, out of the following

Initial values of weights to be learnt.

- Q#83) What are correct facts about SGD Regressor?
- ✓ More iterations require more computation (time).
 - ✓ Tol (tolerance) parameter restricts the # iterations per round.
 - ✓ Training error might actually increase while performing SGD iterations.
- SGD Regressor implements regression using SGD process

(W9) Q#84) Which API will generate a new feature matrix consisting of all polynomial combinations of features ~~less~~ with degrees less than or equal to a specified degree?

PolynomialFeatures

Q#85) What will the following code implement?

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.pipeline import Pipeline

poly_model = Pipeline([('poly-transformation', PolynomialFeatures(
    degree=3)),
    ('linear-regression', LinearRegression())])

poly_model.fit(X, y)
```

It will apply polynomial ~~for~~ transformation of degree 3 & then will train the linear regression model.

Q#86) What will the following code implement?

```
from sklearn.preprocessing import PolynomialFeatures
poly_transform = PolynomialFeatures(degree=2, interaction_only = True)
```

Write the transformed feature $\mathbf{X}$.

$[1, x_1, x_2, x_1x_2]$

- * Ridge Regression : l_2 penalty in SGD Regressor
- * Lasso Regression : l_1 penalty in SGD Regressor
regularization

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↓
Least absolute
shrinkage selection
operator.

Q#87) What is the correct way to perform ridge regularization with regularization rate of 0.2?

from sklearn.linear-model import Ridge
ridge = Ridge(alpha=0.2)

OR

from sklearn.linear-model import SGDRegressor
sgd = SGDRegressor(alpha=0.2, penalty='l2')

Q#88) Which penalty is used to perform both lasso & ridge regularization in polynomial regression using SGDRegressor()?
elasticnet

Q#89) What is the keyword used to set regularization type
penalty
keyword for regularization rate? alpha

Q#90) For penalty = 'elasticnet', l_1 -ratio=0.3 implies l_2 -ratio=0.7

Q#91) Which of the following are true about hyperparameter tuning?

- ✓ Hyperparameter is a parameter whose value is to control the learning process.
- ✓ Hyperparameters are parameters that are not directly learnt from within estimators

Q#92) Name two hyperparameter search approaches?

- ✓ GridSearchCV
- ✓ RandomizedSearchCV

Q#92) Which of the following API is used ~~for~~ to find names and current values for θ parameters for a given estimator?

estimator.get_params()

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Q#93) GridSearchCV *

- ✓ It exhaustively generates candidates from a grid of parameter values specified with the param_grid parameter.
- ✓ It evaluates all possible combinations of parameter values when "fitting" on a dataset & retains the best combination.

*
Q#94) Some parameter may result into failure to fit ≥ 5 folds of data. Which keyword is used to avoid such a failure?

error_score = 0

Q#95) Looking at a set of feature values, how to conclude whether the feature got a long tail distribution?

If the mean is significantly different from the median, then it could indicate a skewness caused by a long tail.